UMD SCHOOL OF PUBLIC HEALTH TEACHING DEMONSTRATION

The Binomial Distribution

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Slides available at **thevazquezlab.com/talks**

Today's goals

By the end of class, you should be able to:

1. Recognize when a count follows a binomial distribution.
2. Compute binomial probabilities and the expected count np.
3. Use binomial reasoning to estimate the size of a population that can be hard to count.

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Running example

A clinic tests $n = 4$ patients for flu. Each patient tests positive (success) with chance $p = 0.25$, independently of the others. X = number of positive tests.
(Illustrative numbers.)

The binomial formula

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Mean and variance

$$E(X) = np, \qquad \mathrm{Var}(X) = np(1-p)$$

Clinic: $E(X) = 4 \times 0.25 = 1, \qquad \mathrm{Var}(X) = 4 \times 0.25 \times 0.75 = 0.75$

Each try is a success with chance p, so one try has mean p and variance $p(1-p)$. Add up n independent tries: mean np, variance $np(1-p)$.

The full expression: n choose k

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How many arrangements give exactly $k = 1$ positive test?

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$+\,-\,-\,-$ $\quad -\,+\,-\,-$ $\quad -\,-\,+\,-$ $\quad -\,-\,-\,+$

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$$\binom{4}{1} = \frac{4!}{1!\,3!} = \frac{24}{1 \times 6} = 4$$

$$P(X = 1) = \frac{4!}{1!\,3!}\,(0.25)^1\,(0.75)^3 = 4 \times 0.25 \times 0.422 = 0.422$$

Adding up: the cumulative distribution

The **cumulative distribution function** (CDF) adds the probabilities up to x:

$$F(x) = P(X \leq x) = \sum_{k=0}^{x} \binom{n}{k} p^k (1-p)^{n-k}$$

k	$P(X = k)$	$P(X \leq k)$
0	0.316	
1	0.422	
2	0.211	
3	0.047	
4	0.004	

$n = 4, p = 0.25$

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What is the chance of **at most 1** positive test, $P(X \leq 1)$?

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What is the chance of **at most 1** positive test, $P(X \leq 1)$?

$P(X \leq 1) = 0.316 + 0.422 = 0.738$

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Watch the inequality:

- $P(X < 2) = P(X \leq 1) = 0.738$
- $P(X \geq 2) = 1 - P(X \leq 1) = 0.262$

Check your understanding

Your turn

A clinic tests $n = 5$ patients; each tests positive with chance $p = 0.2$.

1. What are $E(X)$ and $\mathrm{Var}(X)$?
2. What is $P(X = 0)$?
3. What is $P(X \leq 1)$?

Check your understanding

Your turn

A clinic tests $n = 5$ patients; each tests positive with chance $p = 0.2$.

1. What are $E(X)$ and $\mathrm{Var}(X)$?
2. What is $P(X = 0)$?
3. What is $P(X \leq 1)$?

1. $E(X) = 5 \times 0.2 = 1; \quad \mathrm{Var}(X) = 5 \times 0.2 \times 0.8 = 0.8$
2. $P(X = 0) = \binom{5}{0}(0.2)^0(0.8)^5 = 0.328$
3. $P(X \leq 1) = 0.328 + \binom{5}{1}(0.2)^1(0.8)^4 = 0.328 + 0.410 = 0.737$

What if we do not know the total?

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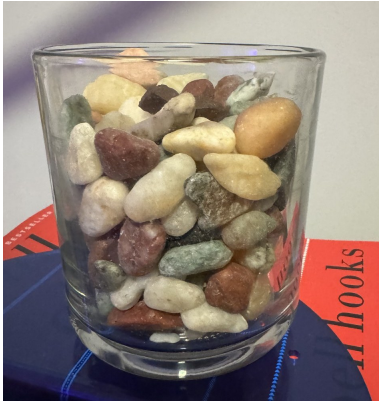
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- Can the binomial help us count a population that nobody can count directly?

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Let’s try it with a jar of pebbles.

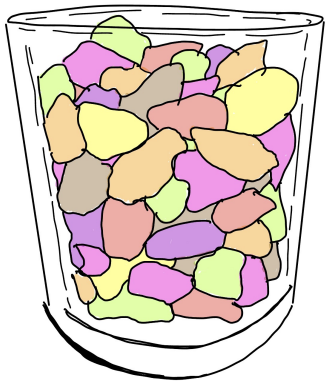
Activity: how many pebbles are in this jar?



Your turn

Without counting every pebble, how could we estimate the total?
Share one idea with your neighbor.

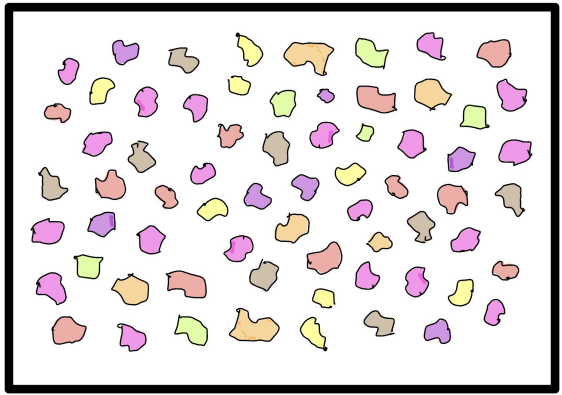
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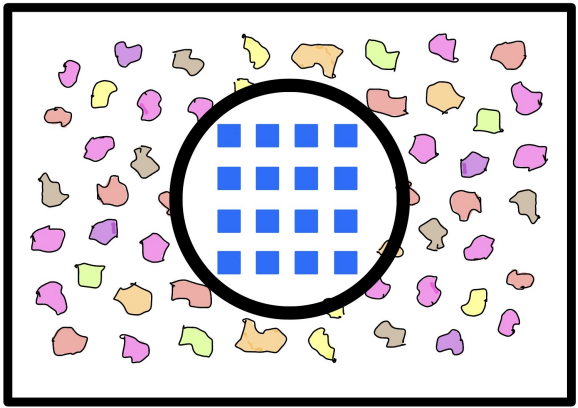
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Our plan, step by step



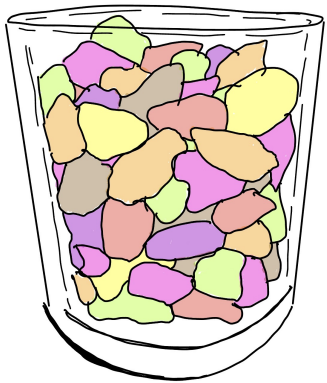
1. Toss the pebbles onto the mat.

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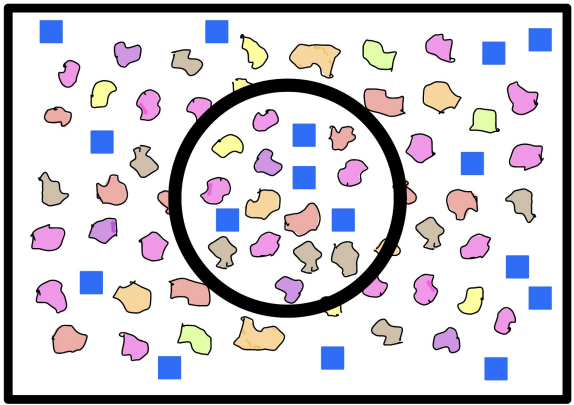
1. Toss the pebbles onto the mat.
2. Swap every pebble inside the circle for a **blue cube**. Count the swaps: M.

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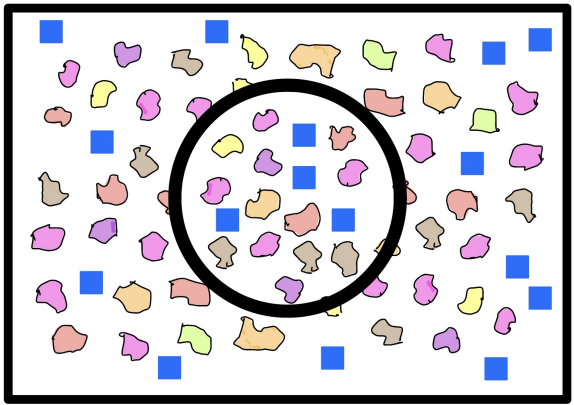
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3. Pour everything back into the jar and mix well.

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Idea: the share of cubes in the circle should be close to the share of cubes in the jar.

The jar is a binomial experiment

On the second toss, each object inside the circle is a cube (success) or a pebble (failure). Swapping keeps the jar's total at N.

In the jar	In the binomial
blue cubes swapped in	M (marked)

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cubes inside the circle	X (successes)
share of the jar that is cubes	$p = M/N$
mixing the jar well	independent tries
cubes and pebbles equally likely to land in the circle	same p on every try

Your turn: how many objects are in the jar?

Your turn

We swapped in $M = 50$ cubes. On the second toss, $n = 40$ objects land in the circle, and $k = 10$ of them are cubes.

About how many objects are in the jar?

Your turn: how many objects are in the jar?

Your turn

We swapped in $M = 50$ cubes. On the second toss, $n = 40$ objects land in the circle, and $k = 10$ of them are cubes.
About how many objects are in the jar?

- In the circle, 10 of the 40 objects are cubes: **one quarter**.

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- In the circle, 10 of the 40 objects are cubes: **one quarter**.
- If the jar is well mixed, cubes should be about one quarter of the whole jar too:

$$\frac{\text{cubes in the circle}}{\text{objects in the circle}} = \frac{\text{cubes in the jar}}{\text{objects in the jar}} \quad \Longrightarrow \quad \frac{10}{40} = \frac{50}{N}$$

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- So 50 cubes are one quarter of the jar: $N \approx 50 \times 4 = 200$.

The binomial gives the same answer

On the second toss, the number of cubes in the circle is binomial with $n = 40$ tries and $p = M/N$. Set the expected count equal to what we saw:

$$E(X) = n \times \frac{M}{N} = k$$

- M = cubes swapped in on the first toss
- n = objects inside the circle on the second toss
- k = cubes inside the circle on the second toss

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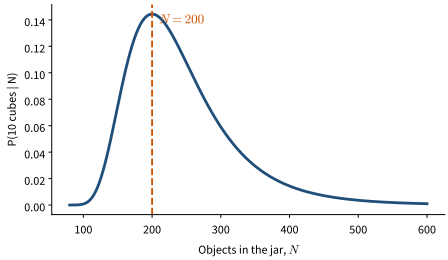
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- M = cubes swapped in on the first toss
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- k = cubes inside the circle on the second toss

The **Lincoln–Petersen estimator**: the same ratio as before. Of all values of N, 200 makes the observed 10 cubes most likely.



Our jar: today's results

$$\hat{N} = \frac{M \times n}{k}$$

Cubes swapped in, M _____

Objects inside the circle, n _____

Cubes inside the circle, k _____

Our estimate, $\hat{N} = Mn/k$ _____

True number in the jar _____

Fill in on the board during class.

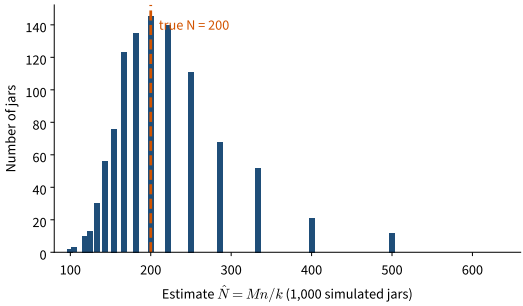
A small simulation: is 200 a lucky answer?

Let the computer fill a jar with exactly **200** objects, swap in 50 cubes, take 40 objects, and compute $\hat{N}$. Repeat 1,000 times.

Your turn

Before you look: will the estimates pile up around 200? Will they be symmetric?

A small simulation: is 200 a lucky answer?



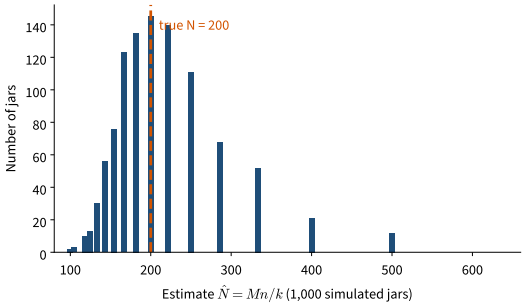
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- Centered near the truth: median 200.

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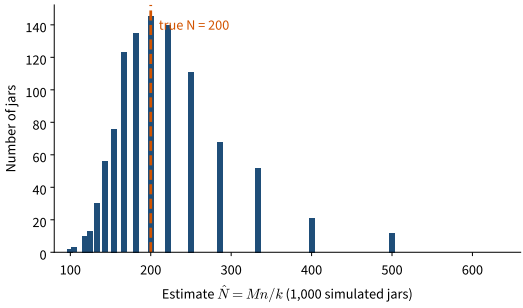
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Your turn

Before you look: will the estimates pile up around 200? Will they be symmetric?

- Centered near the truth: median 200.
- Wide: the middle 90% runs from 133 to 333.

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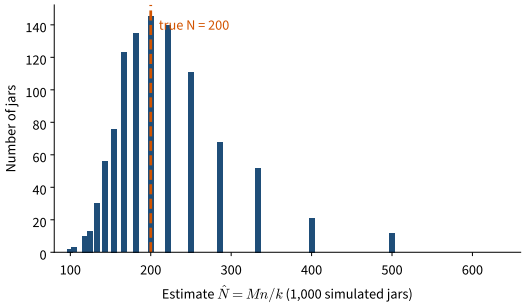
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- Skewed right: a toss with few cubes in the circle gives a very large $\hat{N}$.

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- Centered near the truth: median 200.
- Wide: the middle 90% runs from 133 to 333.
- Skewed right: a toss with few cubes in the circle gives a very large $\hat{N}$.
- Gaps: k is a whole number, so $\hat{N} = 2000/k$ takes only certain values.

My jar at home versus the computer

FIGURE: histogram of your 20 home estimates
next to the 1,000 simulated ones
(send the 20 numbers to Claude)

I repeated the jar experiment **20** times at home.

- Median estimate: _____
- True number: _____

Watch all 20 runs

From the jar to public health

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- Across Latin America, many people with HD are never diagnosed or never reach a specialist. Many countries have no complete registry.



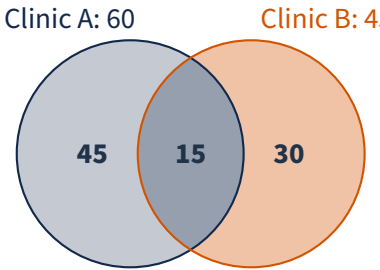
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- Across Latin America, many people with HD are never diagnosed or never reach a specialist. Many countries have no complete registry.
- A health ministry needs the count to plan neurologists, genetic counselors, and long-term care.

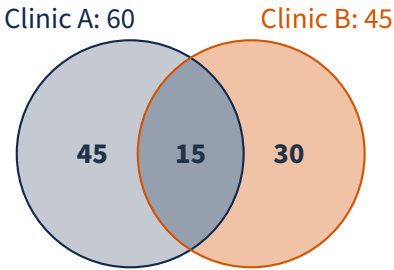


Two clinics, two lists



(Illustrative numbers.)

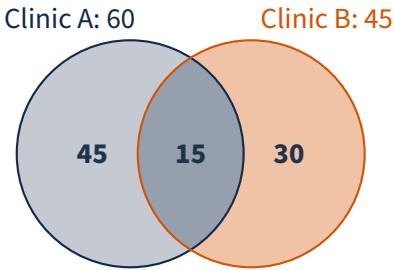
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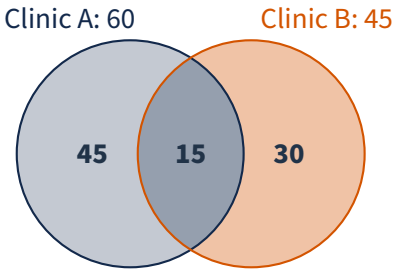
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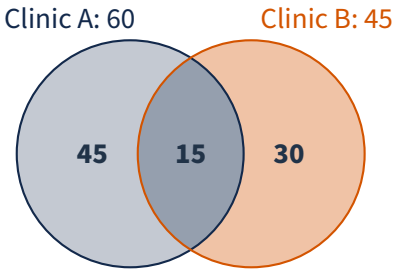
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Your turn

Estimate the number of people with HD in the region.
How many are on neither list?

Answer: the same formula as the jar

In the jar	In the clinics	
M (cubes)	on A's list	60
n (in circle)	on B's list	45
k (cubes in circle)	on both	15

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Why: the share of Clinic B's patients already on Clinic A's list estimates the share of everyone that Clinic A has found:

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- Known to at least one clinic: $60 + 45 - 15 = 90$ (subtract the 15 so nobody is counted twice).
- Estimated on neither list: $180 - 90 = 90$, **half** of everyone with HD.

When the estimate goes wrong

Each binomial condition has a real-world version. Is $\hat{N}$ too large or too small?

Scenario	Condition that fails	What happens
Referral	Independent lists	Clinic A refers its patients to Clinic B, so the overlap is too large and $\hat{N}$ is .

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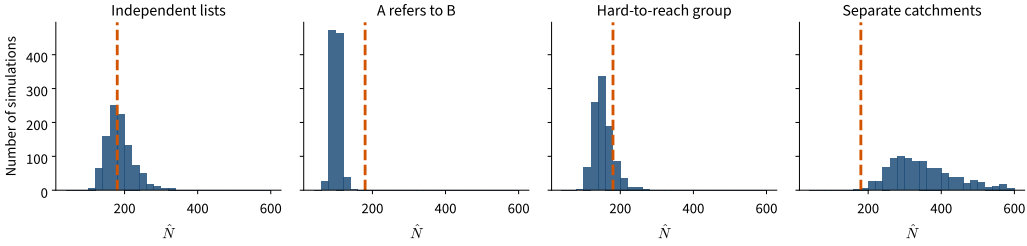
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Two more ways to fail: people die or move between the two lists, and misspelled names hide true matches (k too small, so $\hat{N}$ too large).

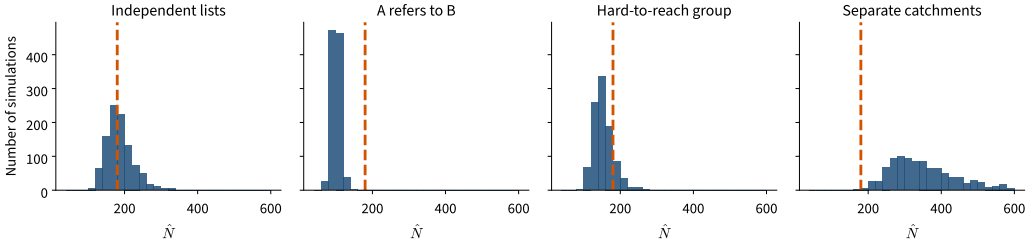
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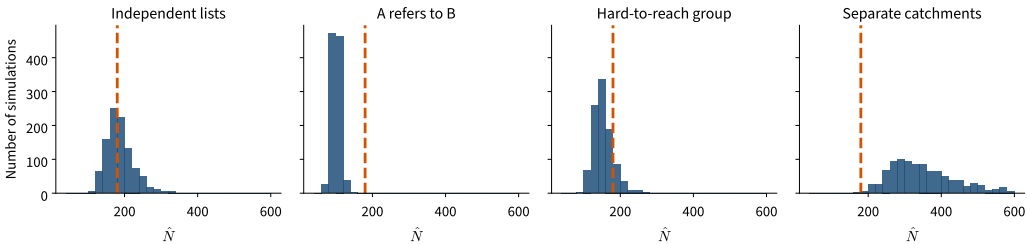
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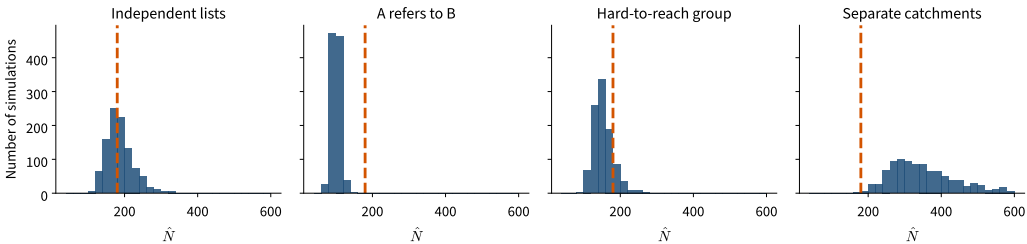


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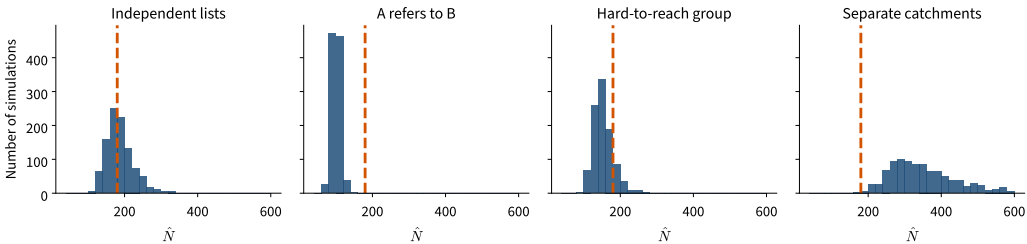
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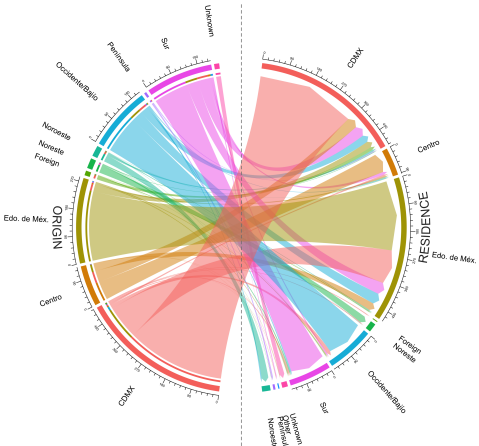
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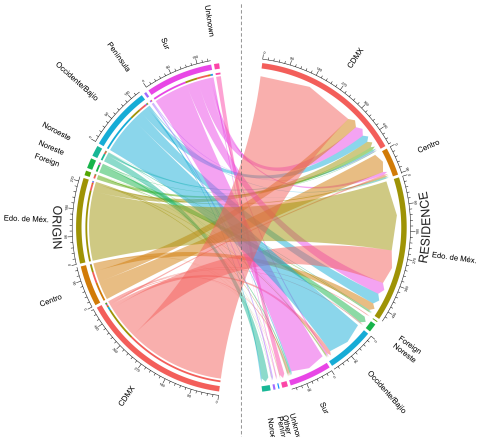
Where do people with HD in Mexico end up living?



Ochoa-Morales A, Vazquez JE, et al. Clinical, molecular, and survival characteristics of Huntington's disease in Mexico: a 30-year cohort from the National Referral Center. In preparation.

- Left: region of origin. Right: region of residence. Many people from other regions now live in Mexico City (CDMX) or Estado de México.

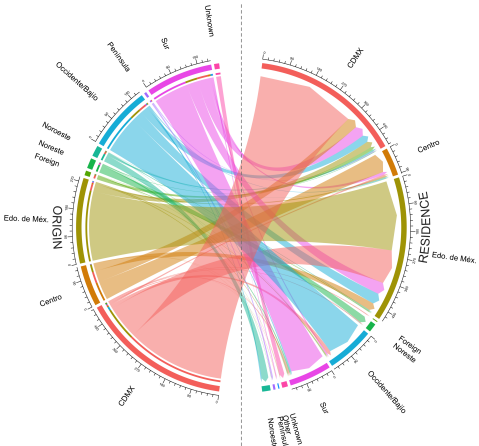
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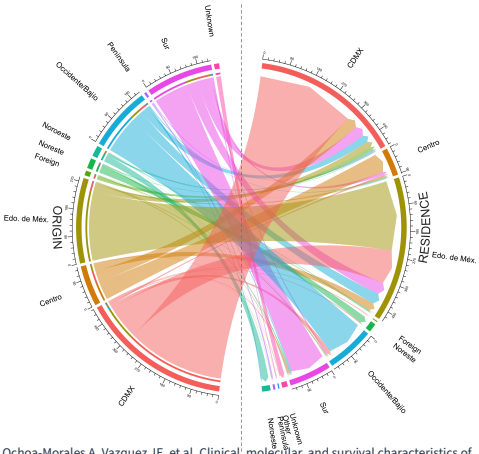
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Your turn

This figure only includes people who reached care. Who is missing?

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- Linking two clinic lists requires privacy protections and, often, consent.

Your turn

Which families would both clinic lists most likely miss?
What would you do to reach them?

In R

```
dbinom(10, size = 40, prob = 0.25)   # P(X = 10)  = 0.14
pbinom(5,  size = 40, prob = 0.25)   # P(X <= 5)  = 0.04
choose(4, 1)                         # 4!/(1! 3!) = 4
pbinom(1,  size = 4,  prob = 0.25)   # P(X <= 1)  = 0.74

M <- 60; n <- 45; k <- 15            # two clinic lists
M * n / k                            # Lincoln-Petersen: 180
(M + 1) * (n + 1) / (k + 1) - 1      # Chapman: 174

# simulation: 1,000 jars with N = 200, M = 50, n = 40
k_sim <- rbinom(1000, size = 40, prob = 50 / 200)
N_hat <- 50 * 40 / k_sim
summary(N_hat); hist(N_hat)
```

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2. Setting np equal to the observed count turns the binomial into an estimator of a population size: $\hat{N} = Mn/k$.
3. The estimate is only as good as the conditions; ask who has no chance of appearing on any list.

Thank you!

Questions?

Jesus E. Vazquez, PhD
Department of Biostatistics
Johns Hopkins Bloomberg School of Public Health

Slides available at **thevazquezlab.com/talks**

Supplementary materials

Want to read more?

The jar experiment: video of all 20 runs

Capture-recapture in epidemiology

- Hook EB, Regal RR (1995). Capture-recapture methods in epidemiology: methods and limitations. *Epidemiologic Reviews* 17(2):243–264. doi:10.1093/oxfordjournals.epirev.a036192
- International Working Group for Disease Monitoring and Forecasting (1995). Capture-recapture and multiple-record systems estimation I: history and theoretical development. *American Journal of Epidemiology* 142(10):1047–1058. PubMed 7485050

Huntington disease and the Lake Maracaibo families

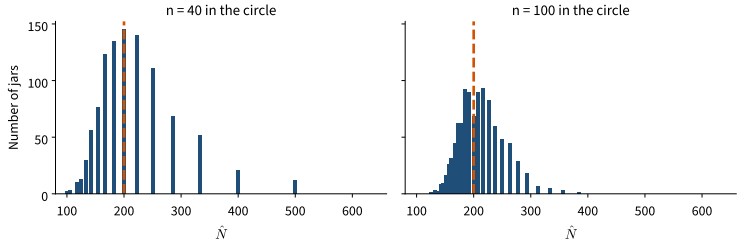
- Gusella JF, et al. (1983). A polymorphic DNA marker genetically linked to Huntington's disease. *Nature* 306:234–238. doi:10.1038/306234a0
- The Huntington's Disease Collaborative Research Group (1993). A novel gene containing a trinucleotide repeat that is expanded and unstable on Huntington's disease chromosomes. *Cell* 72(6):971–983. doi:10.1016/0092-8674(93)90585-E
- Wexler NS, et al. (2004). Venezuelan kindreds reveal that genetic and environmental factors modulate Huntington's disease age of onset. *PNAS* 101(10):3498–3503. doi:10.1073/pnas.0308679101
- Rawlins MD, et al. (2016). The prevalence of Huntington's disease. *Neuroepidemiology* 46(2):144–153. doi:10.1159/000443738

For more depth

For MPH and MS students

- With small overlaps, the **Chapman estimator** $\tilde{N} = \frac{(M+1)(n+1)}{k+1} - 1$ has less bias (derived under the hypergeometric distribution): here $\tilde{N} = 174$.
- With three or more lists, log-linear models for the overlap counts (Poisson distribution) allow some dependence between lists (EPIB 652).
- Public health researchers use the same method to estimate the number of people who inject drugs or undercounted disease cases. For example, see Allen ST, et al. (2019). Estimating the number of people who inject drugs in a rural county in Appalachia. *American Journal of Public Health* 109(3):445–450. doi:10.2105/AJPH.2018.304873

Simulation: what does a bigger second toss buy?



	$n = 40$	$n = 100$
Estimates between 150 and 250	73%	87%
Middle 90% of estimates	133 to 333	156 to 278

More objects in the circle $\Rightarrow$ more cubes found $\Rightarrow$ a less variable $\hat{N}$.